Unsupervised Deep Sparse Features Extraction for SAR Image Segmentation

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***Abstract*— Deep learning (DL) methods usually need to collect a large amount of labeled data to extract deep features. However, due to the difficulty of obtaining numerous labeled data from synthetic aperture radar (SAR) images, unsupervised feature learning has been focused on SAR image processing. In this article, we propose a 3-D sparse model (3-DSM) to extract deep sparse features from SAR images in an unsupervised way. Concretely, 3-DSM learns the convolution kernels by minimizing the error between the features and the constructed sparse maps, without labeled samples. Thus, the discriminative features can be extracted in an unsupervised way by the learned convolution kernels and are able to capture the main structural information of SAR images. Furthermore, to the best of our knowledge, 3-DSM first specifies the sparsity of convolution kernels, with each convolution kernel exhibiting its independence from the others and the redundancy of convolution kernels being diminishing. It means that each convolution kernel extracts its unique structural features of SAR images. Consequently, in the feature extraction, 3-D sparsities have been specified, including width, height, and depth, with the acquisition of discriminative less-redundant features. The effectiveness of 3-DSM is demonstrated by the feature extraction and segmentation of the simulated and real SAR images.**

***Index Terms*— Convolutional neural network (CNN), synthetic aperture radar (SAR) image, unsupervised segmentation, unsupervised sparse feature extraction.**

I. Introduction

SYNTHETIC aperture radar (SAR) image segmentation has been widely applied in various fields, such as change detection [1], [2], [3], target objection [4], [5], and terrain classification [6]. However, because the SAR image suffers from inherent speckle, SAR image segmentation is recognized as a challenging task, and a lot of work have been studied. The achievements of these works can mainly be divided into two categories: model-based algorithms and feature-based algorithms.

The model-based algorithm involves random field model [3], [7], [8], [9], [10], fuzzy clustering model [11], [12], and active contour model [13], [14]. Among them, the random field model is an important tool for SAR image processing. Markov random field (MRF) segments the images from the perspective of statistics. Under the Bayesian framework, MRF combines the pixelwise gray-level information and the label correlation among neighborhood sites to improve the classification performance [3], [7]. Conditional random field (CRF), belonging to the discriminative framework, defines the posterior distribution over labels as a Gibbs field and captures the dependence of the observed data [8], [9], [10], [15], [16]. However, appropriate statistical distribution needs to be chosen carefully for the random field model. On the other hand, inference is a relatively difficult task.

For decades, feature-based methods are studied widely for SAR image segmentation. The classical methods use handcrafted features to achieve segmentation, including histogram measures, gray-level co-occurrence matrix (GLCM), and Gabor filters. As the important popular features, the histogram measures [8], [16] consist of the statistical characteristics involving mean, standard, skewness, entropy, contrast, energy, correlation, and so on. The GLCM [17], [18], as a kind of prominent statistical texture feature, constructs a distribution matrix of co-occurring intensity patterns at a given offset to extract image texture. These handcrafted features have achieved satisfactory results [17], [18]. However, they need expert knowledge to handcraft appropriate feature representations. Deep learning (DL) can extract rich features automatically and has developed rapidly in SAR image processing. Convolutional neural network (CNN) and its variants [19], [20], [21], [22] utilize several cascaded convolution layers for deep feature extraction and classify the image by fully connected layer. In addition, the auxiliary classifier generative adversarial network (ACGAN) [23] introduces an additional generator for data augmentation [24] and then uses a discriminator consisting of convolution networks to extract features and classify the image. The superior performance of these DL models over traditional methods has been demonstrated in SAR data processing [25], [26], [27], [28].

As well known, DL methods need a large amount of labeled data to learn network parameters. However, labeled data are often limited for SAR images. Then, how to extract effective features without enough samples is crucial to SAR image processing. Autoencoders (AEs) [29], [30], [31], [32] and deep belief networks (DBNs) [33], [34], [35], [36] reconstruct the input information in the output layer. The features are extracted by minimizing the reconstruction error, with the label data being not necessary. This principle is widely used in

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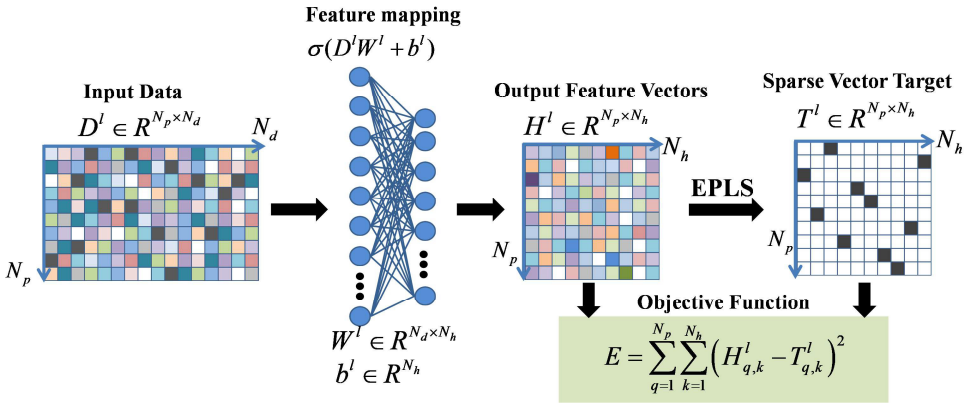

Fig. 1. Parameter learning module of EPLS model in the lth layer. The input data consist of a set of random patches extracted from the output features of the $(l-1)$th layer, and each row represents a random patch. Here, the size of the random patches is also the size of convolution kernels. Then, N_h convolution kernels are applied to the input data for feature mapping, and the output features are H^l. The EPLS algorithm is used to generate sparse target T^l from H^l. Finally, the network parameters are optimized by the objective function, which consists of the error between the output features and the constructed sparse target.

unsupervised deep feature extraction. Although these models can learn feature representation, the basic object of their optimization is to reconstruct the input data rather than learning discriminative features, which limits their performance in the classification task.

Recently, enforcing both population and lifetime sparsity (EPLS) model [37], [38] and its variants [39], [40], [41] are proposed to effectively learn discriminative features in an unsupervised way and achieved promising results in the SAR image classification. In EPLS, a set of convolution kernels are utilized to map each image patch into the feature space, thus forming a feature vector. Then, the EPLS generates the corresponding sparse vectors from these feature vectors, and the model is optimized by minimizing the error between feature vectors and the corresponding sparse ones. The EPLS successfully achieves feature extraction in an unsupervised way.

In fact, in SAR image processing, there are potential improvements for EPLS. First, because the patch-based EPLS model dedicates to extracting a set of sparse feature vectors, the difference of the convolutional kernels is sometimes ignored, and feature maps obtained by the learned convolution kernel sets sometimes exhibit redundancies, which may not be comprehensive for SAR images with complex scene. Furthermore, due to the redundancies of features, a large number of convolution kernels are needed to extract discriminative features in a more complete way, which may introduce massive computation.

Motivated by the discussion above, we propose a 3-D sparse model (3-DSM) to extract expressive features automatically in this article. 3-DSM contains the following contributions.

1) To capture the structural information of SAR image effectively, 3-DSM specifies the sparsity of convolution kernels in the feature extraction. Each convolution kernel is unique and extracts its corresponding structural features. Consequently, 3-DSM is able to capture structural information in a more complete manner and reduces the redundancy of the convolution kernels in the meantime.
2) The sparse convolution kernels are learned by minimizing the error between the features and the constructed 3-D sparse maps. With the sparse convolution kernels, discriminative features are extracted, each of which captures its unique structural information. The sparse features simplify and improve the representation of high-resolution SAR images with complex scene and facilitate the subsequent classification.

The structure of this article is organized as follows. Section II gives a brief review of the EPLS model. The concrete 3-DSM is presented in Section III. Section IV shows the experimental results and analysis. The conclusion is given in Section V.

II. EPLS Model

The EPLS model is a good unsupervised deep features extractor and has achieved promising results in SAR image processing [38], [40], [41]. Three main modules, parameter training module, feature extraction module, and classification module, are involved when applying the EPLS model to SAR image segmentation.

Among them, the parameter training module is the most important part, and the detailed structure of the lth layer is shown in Fig. 1. In this section, the feature mapping of EPLS is given by [37]

$$H^l = \sigma\left(D^l W^l + b^l\right) \tag{1}$$

where $D^l \in R^{N_p \times N_d}$ is the input, N_p is the number of random patches, N_d is the dimension of each patch, $\{W^l \in R^{N_d \times N_h}, b^l \in R^{N_h}\}$ is the network parameters that need to be learned, and $H^l \in R^{N_p \times N_h}$ is the mapping output. The final objective function is expressed as

$$E = \frac{1}{2} \sum_{q=1}^{N_p} \sum_{j=1}^{N_h} \left(H_{q,j}^l - T_{q,j}^l\right)^2 \tag{2}$$

where $T^l \in R^{N_p \times N_h}$ is the sparse target composed of the sparse vector $T_q^l \in R^{N_h}$.

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The EPLS model builds the sparse targets under the constraint of population and lifetime sparsity (LS) based on the feature matrix. The convolution parameters are learned through the objective function consisting only of feature matrixes and the corresponding sparse targets, with the labels being not necessary. However, the sparse target composed of sparse vectors is to guide the learning of a set of sparse feature vectors, while the spatial information of features is ignored. The convolution kernels learned in this way are sometimes redundant, which may not be able to cope with SAR images with the complex scene well.

III. 3-D Sparse Model

As discussed above, the object of EPLS model is to achieve a set of sparse feature vectors, with the independency of convolutional kernels being ignored. Thus, the extracted features of EPLS may be redundant. Consequently, for EPLS, lots of convolution kernels are required to extract features in practice.

In this article, we propose a 3-D sparse model (3-DSM). 3-DSM is able to learn convolution kernels in an unsupervised way and achieve effective feature extraction with the spatial information being considered. The key point of 3-DSM is how to learn the convolution kernels. The learning strategy will be discussed in detail as follows.

Following the principle of EPLS, 3-DSM learns each set of convolution kernels by the constructed 3-D sparse target and thus achieves the feature extraction. Concretely, in the learning stage, besides the sparsity of each feature map, 3-DSM specifies the sparsity of convolution kernels, which is the main contribution of this unsupervised feature representation method. In this way, each convolution kernel is unique and extracts its corresponding structural features. In the 3-DSM, the redundancy of the convolution kernels can be reduced. Also, for SAR images with complex scene, the 3-DSM can extract features in a more complete manner with several convolution kernels compared to EPLS.

A. Definition and Generation of the 3-D Sparse Map Target

In order to learn the convolution kernels in an unsupervised way, the sparse target plays an important role in optimization. 3-DSM defines the sparse targets based on complete feature maps, and the spatial information is considered. Concretely, the sparse target is only composed of 1 and 0 representing activation and inactivation, respectively. The vectors at each location of the sparse map target are one-hot vectors, and the activation is distributed as evenly as possible on the sparse map target, as shown in Fig. 2. Therefore, on the one hand, the sparse map target has good sparsity. Compared to single-dimensional sparse vectors of EPLS model, the sparsity of the sparse map target is reflected in the three dimensions, including width, height, and depth, which is more suitable for convolution networks and promotes the learning of the sparser convolution kernels. Second, N_h maps of a sparse map target are orthogonal to each other. Thus, the biggest differences lie between these maps, which facilitate the learning of independent convolution kernels in an explicit way. Overall, as the optimization target, the 3-D sparse maps can not only promote the learning of sparse convolution kernels but also reduce the redundancy of convolution kernels.

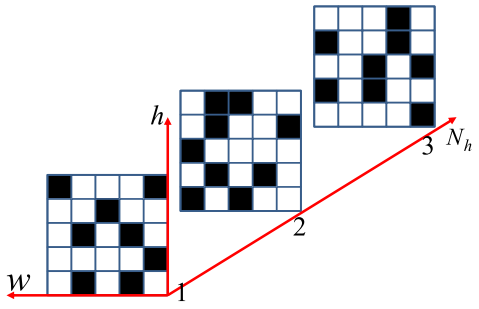

Fig. 2. Sparse map target with the depth N_h being 3. In the figure, the black and white squares represent activation 1 and inactivation 0, respectively. Each sparse map corresponds to the sparse result of a feature map convolved with a convolution kernel. Each map of a sparse target is sparse, and the sparsity of the sparse target is also reflected in depth connecting different maps.

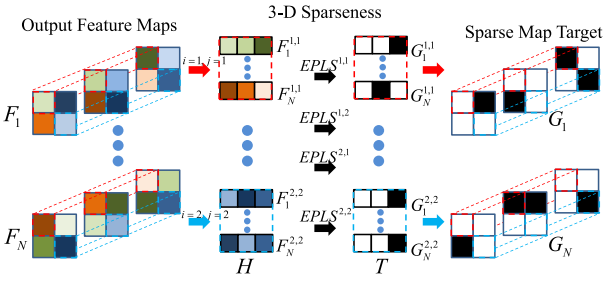

Fig. 3. Example of the process of 3-D sparseness. In this simple example, the width and height of each feature map are set to 2, the depth of the features N_h is set to 3, and the number of the samples is set to N. There are some dotted boxes in feature maps and sparse maps, where the dotted boxes with different colors represent the feature vectors of different locations. The feature vector of a sample at a position is represented by the same color, and the darker the color, the larger the value. Due to the control of the inhibitor a, the activation on the sparse target may not be the maximum item in each feature vector.

To construct the sparse map target for optimization, the 3-DSM process features in a pointwise way. The implementation details are listed in Algorithm 1. We will illustrate the algorithm concretely coupled with a simple example shown in Fig. 3. The first input is a set of feature maps $O = \{F_n \in R^{w\times h\times N_h}\}_{n=1}^{N}$, where w, h, and N_h represent the width, height, and channel number of the feature maps, respectively, and N denotes the number of samples. Another input of the algorithm is the inhibitor vector $a \in R^{N_h}$. The output is a set of sparse maps $S = \{G_n \in R^{w\times h\times N_h}\}_{n=1}^{N}$ and the inhibitor vector a. First, 3-DSM puts the 3-D features into 2-D space, and the process is shown in Fig. 3. Concretely, as listed in Algorithm 1, the feature $O = \{F_n\}_{n=1}^{N}$ is transformed into a matrix $H \in R^{N_s\times N_h}$ in a pointwise way (line 3 in Algorithm 1), where $N_s = w \times h \times N$ indicates the number of feature vectors of N samples (or features). Then, we invoke the EPLS algorithm to compute the 2-D sparse matrix $T \in R^{N_s\times N_h}$. In particular, for each sample, the index x of the maximum value after subtracting the inhibitor a from the feature vector h is set as activation 1 (lines 7 and 8). The inhibitor vector is used to balance the number of times each dimension is selected and progressively increases its value in index x by

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N_h/N_s (line 9). In this way, the activation is distributed as evenly as possible across the depth, width, and height. Finally, the inverse operation is performed to transform the 2-D sparse matrix T into the 3-D sparse maps $S = \{G_n\}_{n=1}^{N}$ (line 11).

Algorithm 1 3-D Sparseness

Iutput: $O = \{F_n \in R^{w\times h\times N_h}\}_{n=1}^{N}$, N, a
Output: $S = \{G_n \in R^{w\times h\times N_h}\}_{n=1}^{N}$, a
1: **for** $u = 1 : w$ **do**
2: **for** $v = 1 : h$ **do**
3: $H = \{F_n^{u,v}\}_{n=1}^{N}$, $T = 0$
4: $H = (H - \min(H))/(\max(H) - \min(H))$
5: **for** $n = 1 \to N_b$ **do**
6: $h = H(n, :)$
7: $x = \arg\max_j (h - a)$, $\forall j \in \{1, 2, \ldots, N_h\}$
8: $T(n, x) = 1$
9: $a_x = a_x + (N_h/w \times h \times N)$
10: **end for**
11: $\{G_n^{u,v}\}_{n=1}^{N} = T$
12: **end for**
13: **end for**

B. Learning of Sparse Convolution Kernels

With the 3-D sparse maps being the target of optimization, the sparse convolution kernels can be learned in an unsupervised way. Concretely, the network can be optimized by minimizing the error between the feature maps and their corresponding sparse map target. The difference between different maps of a sparse target promotes the sparsity and independency of the convolution kernels. Thus, each convolution kernel is distinctive and captures its special structural features.

1) Backbone of 3-DSM: The backbone of 3-DSM in the lth layer is shown in Fig. 4. The input of the model is the samples selectively sampled from SAR images, which is the same as the EPLS model. However, instead of extracting random patches as in the EPLS model, the samples or features are directly fed to the convolution layer to obtain convolution features $F^l \in R^{w_l\times h_l\times N_h}$, and the convolution layer can be expressed as [19]

$$V_j^l = \sum_{i=1}^{m} K_{ji}^l * F_i^{l-1} + b_j^l$$
$$F_j^l = \sigma\left(V_j^l\right) \tag{3}$$

where V_j^l is the convolution sum of m inputs F_i^{l-1} to the jth output feature map of the lth layer, $F^{l-1} \in R^{w_{l-1}\times h_{l-1}\times m}$ is the output features of (l-1)th layer or samples sampled selectively from SAR image in the first layer, $F^l \in R^{w_l\times h_l\times N_h}$ is the output features of lth layer, $\{w_{l-1}, h_{l-1}\}$ and $\{w_l, h_l\}$ are the width and height of the input and output features, respectively, and $\{m, N_h\}$ are the depth of input and output features at the lth layer, respectively. σ is a pointwise nonlinear activation. $\Gamma^l = \{K^l \in R^{k\times k\times m\times N_h}, b^l \in R^{N_h}\}$ are the convolution parameters, convolution kernels, and bias, where k is the width and height of each convolution kernel and $*$ indicates the convolution operation.

Then, the 3-D sparse maps $G^l \in R^{w_l\times h_l\times N_h}$ corresponding to the convolution features F^l are constructed based on the EPLS algorithm in a pointwise way. Notably, the 3-D sparse maps are sparse in multiple dimensions. In detail, each map of a 3-D sparse target is sparse, and different maps of the 3-D sparse map are mutually orthogonal. Finally, the 3-D sparse maps are introduced into the objective function to update the parameters of the next round.

2) Objective Function of 3-DSM: With the 3-D sparse maps being the target, the network can be optimized effectively in an unsupervised way. Concretely, the network parameters are learned by iteratively minimizing the error between the convolution features F^l and the corresponding sparse maps G^l. Also, the objective function can be expressed as

$$E = \frac{1}{2}\sum_{n=1}^{N}\sum_{j=1}^{N_h}\sum_{u=1}^{w_l}\sum_{v=1}^{h_l}\left(F_{v,u,j,n}^l - G_{v,u,j,n}^l\right)^2. \tag{4}$$

3) Learning Process: To reduce the redundancy of the convolution kernels and improve their sparsity, the optimization object of 3-DSM is the complete convolution features rather than a set of feature vectors of EPLS. In detail, the parameters are adjusted iteratively using (4) based on the mini-batch stochastic gradient descent (SGD) algorithm. For G^l, a reasonable assumption is made that G^l is independent of Γ^l, and thus, $(\partial G^l/\partial\Gamma^l) = 0$. According to the chain rule, an "error item" δ_j^l is defined and deduced to simplify the expression

$$\begin{aligned}\delta_j^l &= \frac{\partial E}{\partial V_j^l}\\ &= \frac{\partial E}{\partial F_j^l}\frac{\partial F_j^l}{\partial V_j^l}\\ &= \left(F_j^l - G_j^l\right)\odot\sigma'\left(V_j^l\right)\end{aligned} \tag{5}$$

where $\odot$ denotes the dot product. Thus, the error gradient of the convolution kernels and bias can be calculated as

$$\begin{aligned}\frac{\partial E}{\partial K_{ji}^l} &= \frac{\partial E}{\partial V_j^l}\frac{\partial V_j^l}{\partial K_{ji}^l}\\ &= \delta_j^l * F_i^{l-1}\end{aligned} \tag{6}$$

$$\begin{aligned}\frac{\partial E}{\partial b_j^l} &= \frac{\partial E}{\partial V_j^l}\frac{\partial V_j^l}{\partial b_j^l}\\ &= \delta_j^l.\end{aligned} \tag{7}$$

Finally, the parameters of the model can be updated by

$$\begin{aligned}K_{ji}^l[t+1] &= K_{ji}^l[t] - \eta\delta_j^l * F_i^{l-1}\\ b_j^l[t+1] &= b_j^l[t] - \eta\delta_j^l\end{aligned} \tag{8}$$

where t and $t+1$ are the number of iterations.

C. Unsupervised Feature Extraction Using 3-DSM

The learned convolution kernels are not only sparse but also independent. Furthermore, discriminative sparse features can be extracted, each of which owns its unique structural information. Thus, the features have low redundancy, which is

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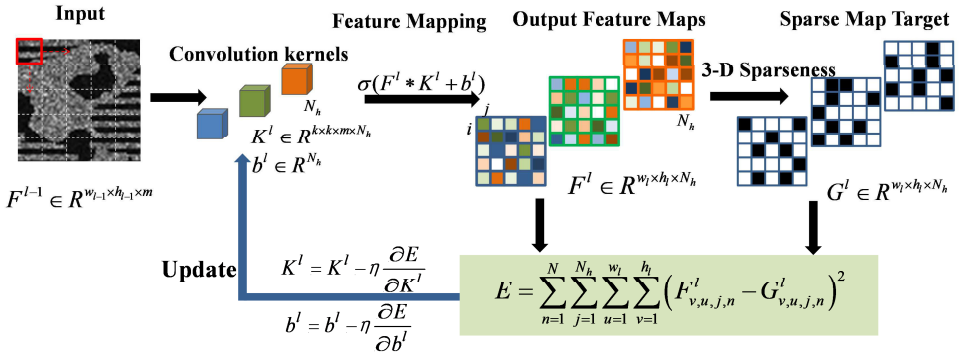

Fig. 4. Backbone of 3-DSM with a single layer. To simplify the figure, the number of convolution kernels N_h is set to 3 here. The feature maps with blue, green, and orange borders are obtained by the convolution kernels of the corresponding colors. The thick black arrow indicates the forward propagation process of information, and the thick blue arrow represents the backward propagation of error.

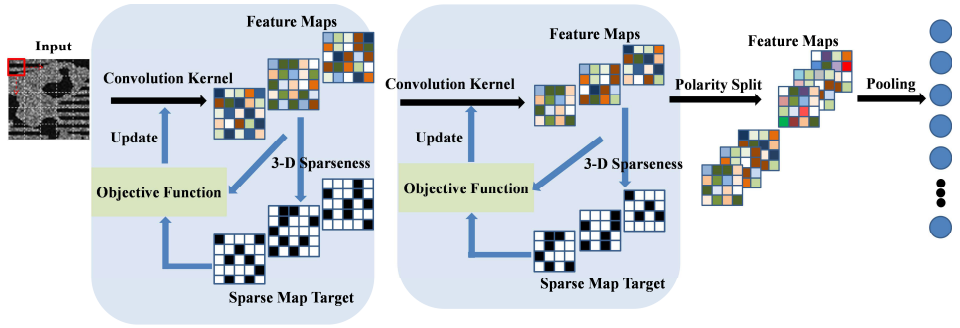

Fig. 5. Complete 3-DSM model for SAR image segmentation. The complete structure includes two convolution layers and a PS. The details of each convolution layer have been described in Fig. 4. PS enriches the features, thus excavating more information. Finally, the extended features are pooled and fed to the classifier.

beneficial for subsequent classification. Consequently, 3-DSM is an unsupervised feature extraction method in a way of optimizing for sparsity. Also, the complete structure of 3-DSM is shown in Fig. 5.

The deep network of 3-DSM is based on the greedy layer-wise (GLW) unsupervised pretraining strategy. Algorithm 2 describes this process using pseudocode. The input of the algorithm is a set of samples $O^0 = \{F_n^0\}_{n=1}^N$ and the number of layers L. The output is the extracted features $O^L = \{F_n^L\}_{n=1}^N$ and the learned convolution parameters Γ. For each layer, the inhibitor vector a is initialized to the zero vector with its length being N_h (line 5). Then, mini-batches are used to train the network, and F_b^{l-1} is the mini-batch of F^{l-1} with the batch size being N_b. For each batch, the convolution parameters updated in the last iteration are used to obtain the convolution features of this iteration (line 8 in Algorithm 2). The 3-D sparse maps G_b^l, corresponding to the features F_b^l, are obtained by the 3-D sparseness algorithm (lines 9). With the SGD algorithm, the network parameters are gradually updated by minimizing the objective function (4) (lines 10 and 11).

To further improve the performance, the features are extended by the polarity split (PS) [42] method in the last layer of the model to extract more information. The formula of PS is given by

$$\begin{aligned} F_+^L &= \sigma\left(F^{L-1} * K^L + b^L\right) \\ F_-^L &= \sigma\left(F^{L-1} * \left(-K^L\right) + b^L\right) \\ F^L &= \left[F_+^L, F_-^L\right] \end{aligned} \tag{9}$$

Algorithm 2 GLW Unsupervised Pretraining Algorithm

Input: $O^0 = \{F_n^0\}_{n=1}^N$, L
Output: $O^L = \{F_n^0\}_{n=1}^N$ and parameters $\Gamma=\{\Gamma^1, \ldots, \Gamma^L\}$, where $\Gamma^l=\{K^l, b^l\}$ $\forall l \in \{1, 2, \ldots, L\}$

```
1: for l → 1 to L do
2:     Γ=small random values
3:     repeat
4:         Shuffle F^{l-1} randomly
5:         a = 0
6:         for b = 1 → ⌊N/N_b⌋ do
7:             F_b^l = σ(F_b^{l-1} * K^l + b^l)
8:             [G_b^l, a] = 3-D Sparseness(F_b^l, N_b, a)
9:             Calculate error by (4)
10:            Update parameters by (8)
11:        end for
12:    until stop condition verified
13:    Normalize feature maps
14: end for
```

where F_+^L is the positive component, F_-^L is the negative component, and F^L represents the concatenation of the two components in the depth dimension.

In short, 3-DSM is an effective unsupervised deep sparse feature extraction method. The model is optimized by minimizing the error between the feature maps and the constructed 3-D sparse maps. Sparse convolution kernels are learned in an

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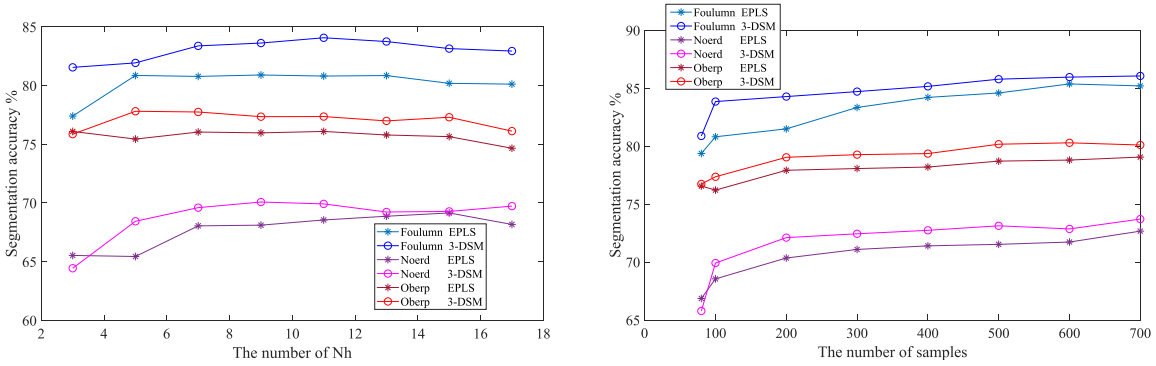

Fig. 6. Influence of the number of convolution kernels and samples on the segmentation accuracies of different SAR images. (a) Relationship between the segmentation accuracy and the number of convolution kernels in the three regions. (b) Relationship between the segmentation accuracy and the number of samples in the three regions.

TABLE I
PRIOR INFORMATION ABOUT THE REAL SAR IMAGES

Image	Size	Class	Band	Source	Location
Fig.14	700×700	3	C	GaoFen-3	Daxing
Fig.15	512×512	5	L	ESAR	Oberpfaffenhofen
Fig.16	400×400	4	X	TerraSAR	Noerdlinger
Fig.17	300×300	4	L	EMISAR	Foulumn

explicit way. Each convolution kernel is independent of the others and has more perspective for feature extraction, which reduces the redundancy of the convolution kernels. In this way, each feature map captures specific structural information, and diverse sparse features are extracted. The sparse features simplify and improve the representation of the input and facilitate the subsequent classification.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

Simulated data and four real SAR data from GaoFen-3, ESAR, TerraSAR, and EMISAR are used to validate the effectiveness practicality of our model. The prior information about the real SAR images is listed in Table I. The experiment results and discussion will be presented later.

A. Experiment Settings and Hyperparameter Selection

Some experiment parameters of 3-DSM are set here. The samples with the size being 11×11 [25] are selected randomly in each category from the original SAR image. The initial values of the convolution kernel with size being 3×3 [43] are randomly drawn from (0, 0.1), and the final convolution kernels are obtained after 200 iterations with the learning rate being 0.001. The number of the network layers and the nonlinear activation are set to 2 [38], [41] and rectified linear unit (ReLU) function. Considering the time of parameter learning, the size of the mini-batch is set to 10. Finally, the deep features at the last layer processed by PS are pooled into four quadrants by average pooling [44], which are fed to support vector machine (SVM) [45] for classification. The experiment is performed on MATLAB, Core-i7 4.2 GHz.

Besides, the number of convolution kernels and the number of samples are two important hyperparameters. We conduct experiments on three real SAR images with different scenes to analyze the influence of the parameters on segmentation accuracy. The three real SAR images are shown in Figs. 15(a)–17(a). Also, the experimental results are shown in Fig. 6(a) and (b). As the number of convolution kernels increases, the amount of computation will increase. However, the segmentation accuracy of the model with a small number of convolution kernels is not good enough. Therefore, as shown in Fig. 6(a), we can see that 11 is an appropriate number of convolution kernels for our model. The samples are used to train the SVM classifier. Considering the difficulty of obtaining labeled data in SAR images, we hope to achieve good performance with a small number of samples. Thus, the number of samples is set as 200 per class according to Fig. 6(b).

In terms of time complexity, the 3-DSM model needs to compute F and G in each layer, whose costs are $\mathcal{O}(w_l \times h_l \times k^2 \times m \times N_h)$ and $\mathcal{O}(N \times w_l \times h_l \times N_h)$, respectively. The corresponding space complexity is $\mathcal{O}(k^2 \times m \times N_h)$.

B. Reference Models

1) DBN [33]: Similar to 3-DSM, DBN as an unsupervised model is trained in the GLW method. In our experiment, the numbers of hidden units of each layer are 90 and 80. The nonlinear activation is the sigmoid function; 200 samples for each class are used to train the SVM classifier. The time complexity is $\mathcal{O}(w_{l-1} \times h_{l-1} \times m \times N_{out} \times (2N_G + 1))$, where N_{out} represents the dimensionality of the output features and $N_G = 1$ denotes the number of iterations of sampling. Also, its space complexity is $\mathcal{O}(w_{l-1} \times h_{l-1} \times m \times N_{\text{out}})$.

2) SAE [29]: Similar to 3-DSM, sparse AE learns the network parameters in the GLW unsupervised pretraining fashion. It is a classical and widely used feature extraction method. The numbers of hidden units of each layer are 100 and 90. The nonlinear activation is the sigmoid function. The sparsity and the sparsity penalty are set to 0.05 and 0.001, respectively; 200 samples for each class are used to train the SVM classifier. The time complexity is $\mathcal{O}(w_{l-1} \times h_{l-1} \times m \times N_{\text{out}})$ and the space complexity is $\mathcal{O}(w_{l-1} \times h_{l-1} \times m \times N_{\text{out}})$.

3) EPLS Model [37], [38]: The model provides a new strategy for unsupervised feature learning in a GLW way by

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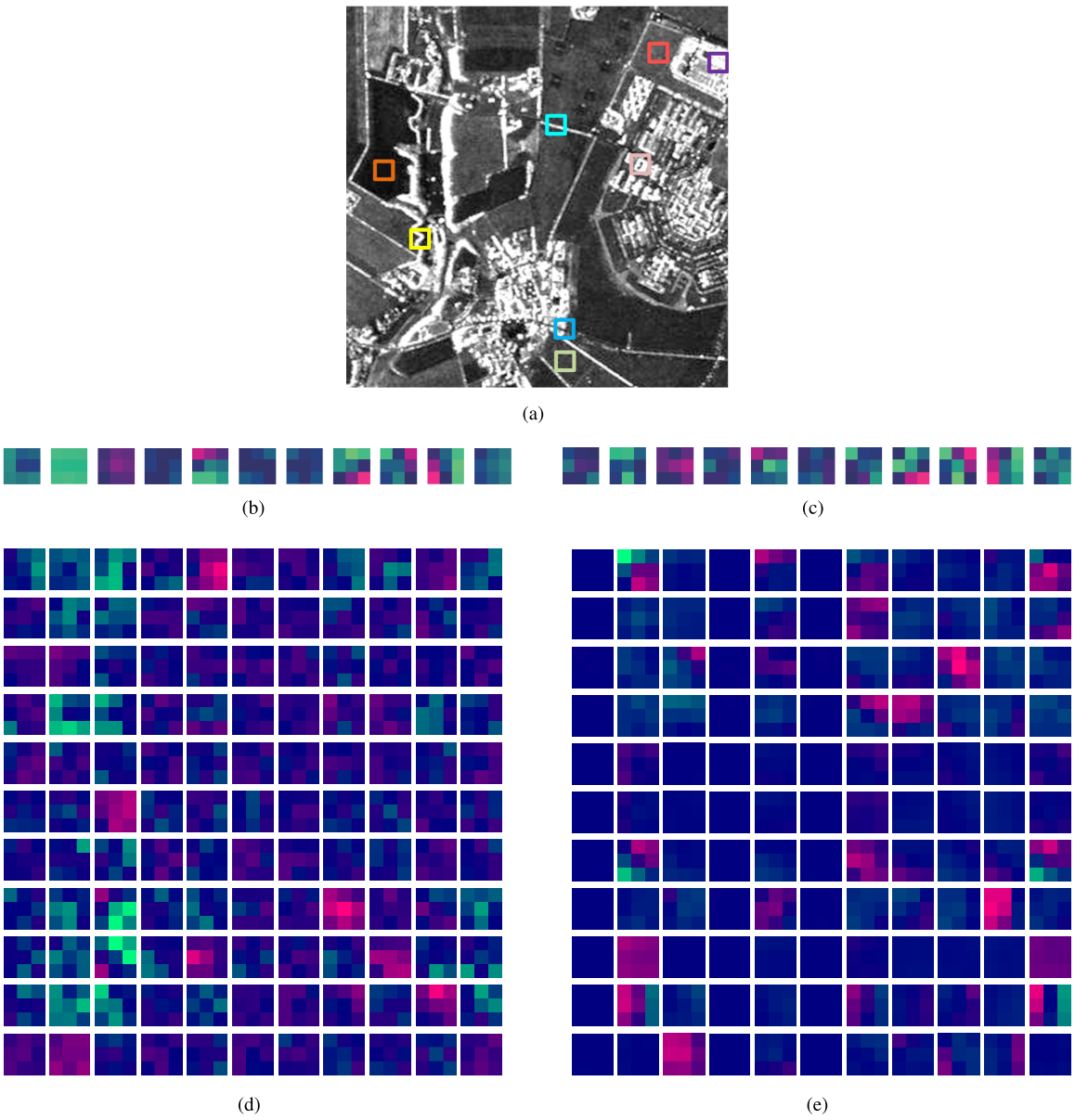

Fig. 7. Convolution kernels of 3-DSM and EPLS model. (a) Real SAR image. (b) Convolution kernels of EPLS model for the first layer. (c) Convolution kernels of 3-DSM for the first layer. (d) Convolution kernels of EPLS model for the second layer. (e) Convolution kernels of 3-DSM for the second layer. In (b)–(e), the blue indicates 0, the positive values are shown in red, and the negative values are shown in green. The brighter the color, the bigger the absolute value. In (d) and (e), each row represents a convolution kernel, and each column represents the depth of the convolution kernels.

constructing a sparse target. In our experiment, the model has two layers, and the number of patches and convolution kernels in each layer is 10000 and 11, respectively. The nonlinear activation is set to the ReLU; 200 samples for each class are used to train the SVM classifier. To improve the performance, the features at the last layer are extended by PS and then pooled into four quadrants by average pooling. The time complexity calculating the feature vectors H and the sparse vector target T is $O(k^2 \times m \times N_h)$ and $O(N_p \times N_h)$, respectively, and the space complexity is $\mathcal{O}(k^2 \times m \times N_h)$.

4) ACGAN [23]: Compared with the traditional generative adversarial network (GAN), ACGAN can not only achieve data augmentation with the generator but also introduce an additional classifier in the discriminator for label prediction. Thus, it has advantages for classification tasks with a small number of samples. In our experiment, the discriminator consists of two convolution layers with the number of convolution kernels being 16 and 32. LeakyReLU is chosen as nonlinear activation, and the rate of dropout is set to 0.25; 200 samples per class are used to train the entire network. The time and space complexity of the convolution layer are $\mathcal{O}(w_l \times h_l \times k^2 \times m \times N_h)$ and $\mathcal{O}(k^2 \times m \times N_h)$, respectively, where w_l represents the width of output features of the lth layer. Also, the time and space complexity of the fully connected layer are $\mathcal{O}(w_{l-1} \times h_{l-1} \times m \times N_{\text{out}})$ and $\mathcal{O}(w_{l-1} \times h_{l-1} \times m \times N_{out})$, respectively.

C. Diversity and Sparsity Analysis of 3-DSM' Features

To intuitively illustrate the properties of the convolution kernels learned by 3-DSM, we apply the model to the real SAR image shown in Fig. 7(a). The obtained convolution kernels of 3-DSM [Fig.7(c) and (e)] are compared with that of EPLS

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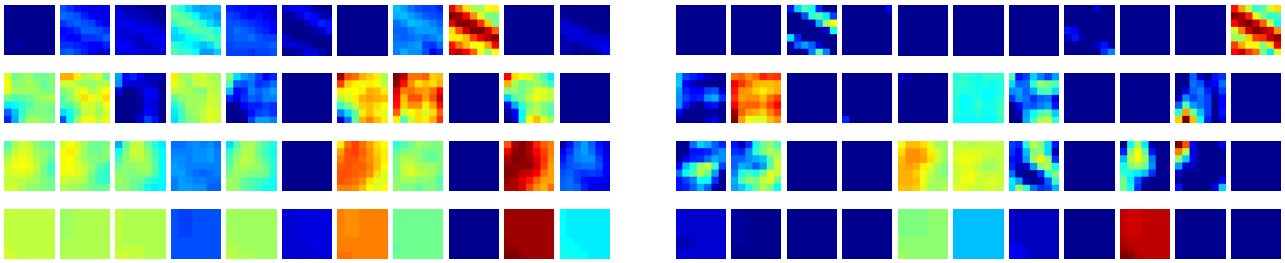
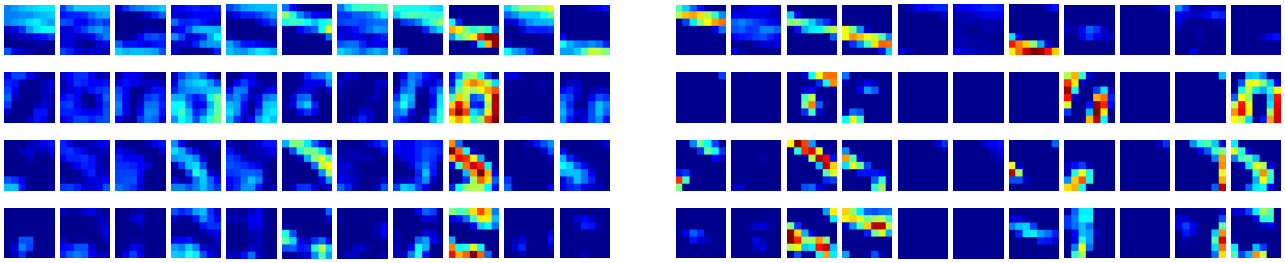


(a) (b)

Fig. 8. Feature maps of four homogeneous regions obtained by EPLS and 3-DSM. (a) Feature maps of EPLS model for the second layer. (b) Feature maps of 3-DSM for the second layer. For (a) and (b), the row represents different regions and the column denotes the number of convolution kernels. The first, second, third, and fourth rows represent the features of the four homogeneous regions framed in purple, red, green, and orange in Fig. 7(a), respectively. The blue denotes zero and the other colors represent nonzero. Compared with the feature maps in (a), the feature maps in (b) are sparser and have low redundancy.



(a) (b)

Fig. 9. Feature maps of four heterogeneous regions obtained by EPLS and 3-DSM. (a) Feature maps of EPLS model for the second layer. (b) Feature maps of 3-DSM for the second layer. The first, second, third, and fourth rows, respectively, represent the feature maps of the four heterogeneous regions framed in cyan, pink, yellow, and blue in Fig. 7(a), respectively. The blue represents zero, and the other colors represent nonzero. Compared with the maps of (a), the maps of (b) are sparser and less redundant.

model [Fig. 7(b) and (d)]. Obviously, there are a lot of zero matrices (the blue square) in Fig. 7(e), such as the first, fourth, and sixth columns and many other locations. However, there are few zero matrixes in Fig. 7(d). Therefore, the convolution kernels learned by 3-DSM are sparser. In 3-DSM, the learned sparse convolution kernels are of significant differences in terms of structure. Concretely, each sparse convolution kernel in Fig. 7(e) exhibits its independence from the others, and the redundancy of the convolution kernels is reduced. In this way, the sparsity of convolution kernels is specified.

Furthermore, using these sparse convolution kernels, we can extract rich features from different perspectives. Following the above experiment, the corresponding feature maps are obtained and shown in Figs. 8 and 9. We take the feature maps of eight representative regions as examples for analysis, including four homogeneous regions in Fig. 8 and four heterogeneous regions in Fig. 9. Intuitively, Fig. 8(b) has more zero matrices (the blue square) than Fig. 8(a), which indicates that the features extracted by 3-DSM are sparser. More importantly, the features of the third and fourth rows in Fig. 8(a) [the green and orange regions in Fig. 7(a)] have many similar feature maps, while the corresponding feature maps in Fig. 8(b) are independent. Thus, the sparse features extracted by 3-DSM are more informative and less redundant. Similarly, for heterogeneous regions with structural information, each feature in Fig. 9(a) has several feature maps with similar structures of different intensity, such as the feature maps of the second, fourth, and ninth columns in the second row [the pink region in Fig. 7(a)]. However, in Fig. 9(b), there are few feature maps with similar structures in each row. This indicates that each nonzero feature map extracted by 3-DSM has its own unique characteristics to predict the class, which reduces the redundancy of the feature maps and promotes the expression ability of the features. In this way, the extracted features are more comprehensive, which is beneficial for the subsequent classification.

D. Ablation Study

To illustrate the effectiveness of our model, 3-DSM without LS and 3-DSM without PS are compared with our proposed 3-DSM in the classification experiment. The experiments can be seen as an ablation study to verify the validity of EPLS and PS strategies.

First, in EPLS, the sparse targets for optimization need to fulfill strong population and LS. To analyze the effectiveness of EPLS, we design an ablation experiment in which the constructed sparse targets do not satisfy LS. To achieve that, we remove the inhibitor vector a in Algorithm 1 that ensures the LS of sparse targets. Then, in line 7 of the 3-D sparseness algorithm, x is the index where the largest feature is located, i.e., $x = \arg\max_j (h_j), \forall j \in \{1, 2, \ldots, N_h\}$. Thus, the constructed sparse targets satisfy only the population sparsity. The experiment is performed on a simulated image and a real SAR image of the Foulumn region, and the results are shown in Fig. 10(c) and Table II. As listed in Table II, without LS, the accuracy of 3-DSM decreases by 27.72% and 28.27% for the simulated and real SAR images, respectively. Population sparsity in sparse targets ensures the extraction of sparse features,

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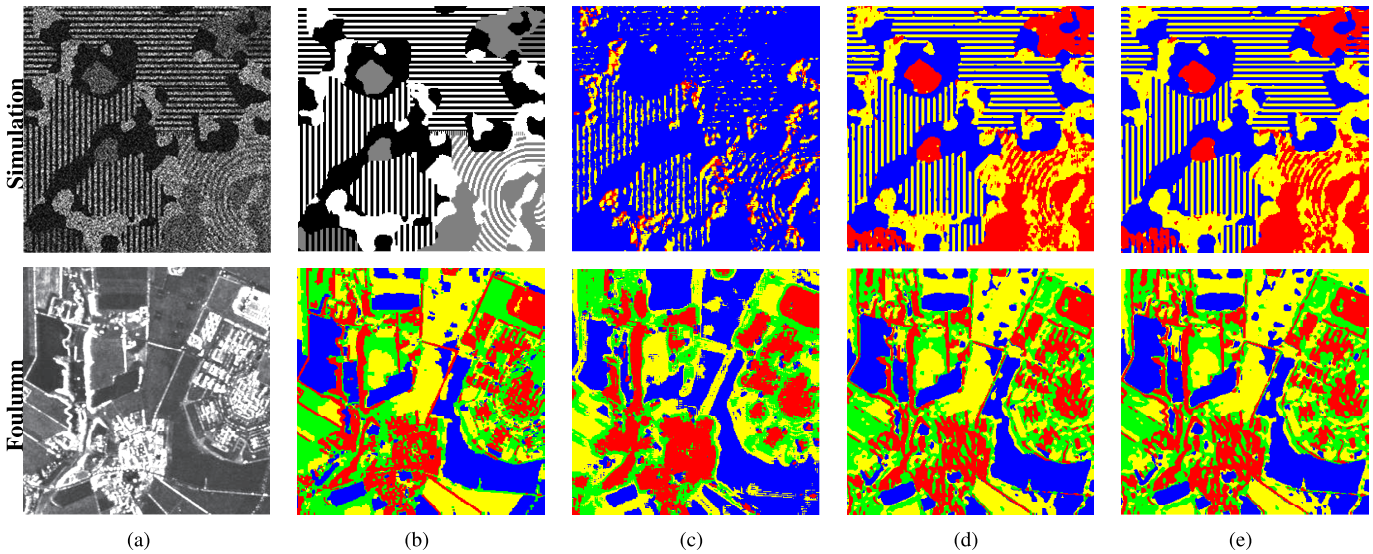

Fig. 10. Segmentation results of a simulated image and a real SAR image of the Foulumn region. (a) Experiment data of a simulated image and a real SAR image of the Foulumn region. (b) Ground truth. (c) Segmentation based on 3-DSM without LS. (d) Segmentation based on 3-DSM without PS. (e) Segmentation based on 3-DSM.

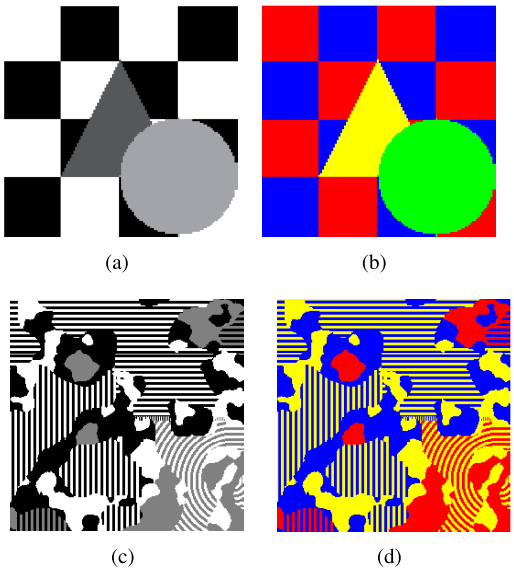

Fig. 11. Ground truth of the simulated images. (a) Ground truth1. (b) Colorful ground truth1. (c) Ground truth2. (d) Colorful ground truth2.

and LS improves the difference between these sparse features. Thus, the sparse features extracted by 3-DSM without LS have low discrimination, leading to many misclassifications, as shown in Fig. 10(c). Compared to 3-DSM without LS, 3-DSM has a stronger ability to capture details, especially for fine structures.

On the other hand, to illustrate the effectiveness of PS, we design another ablation experiment where PS is removed. The features extracted in the last layer are fed to the classifier after pooling, and the experiment results and quantitative evaluation are shown in Fig. 10(d) and Table II. With the introduction of PS, the accuracies of 3-DSM in the simulated and real SAR images are enhanced by 1.62% and 1.05%, respectively. Moreover, the segmentation result of

TABLE II
QUANTITATIVE EVALUATION OF THE ABLATION EXPERIMENTS

	3-DSM without LS	3-DSM without PS	3-DSM
Simulated image	65.04%	91.14%	**92.76%**
Foulumn	55.59%	82.81%	**83.86%**

the simulated image shows that 3-DSM obtains smoother homogeneous regions and preserves more details. Thus, the introduction of PS enriches the feature representation and improves the segmentation accuracy.

E. Experimental Results on the Simulated SAR Images

The simulated SAR images are first used to demonstrate the effectiveness of our model. The first simulated image is obtained by adding multiplicative noise with noise variance being 1.0 to the ground truth of Fig. 11(a), as shown in Fig. 12(a). Similarly, multiplicative noises with noise variance being 1.0, 0.5, and 0.2 are added to the ground truth of Fig. 11(c), and then, a set of simulated images with the corresponding effective numbers of looks (ENL) [46] being 1-look, 2-look, and 5-look, respectively, are obtained, as shown in Fig. 13(a).

As shown in Figs. 12(b) and 13(b), DBN achieves good results in maintaining region homogeneity but has poor performance in edge locations. SSAE captures the fine stripe structures well but is susceptible to multiplicative noise, as shown in Figs. 12(c) and 13(c). The two models attempt to explicitly reconstruct the input distribution for deep feature extraction while ignoring the discrimination of output features. To extract representative deep features in an unsupervised way, the EPLS model constructs the sparse matrixes for optimization and achieves good results shown in Figs. 12(d) and 13(d). Compared with the EPLS model, our model achieves better results in edge location [see Fig. 12(f)] and its advantages are more obvious in the images with complex texture structures

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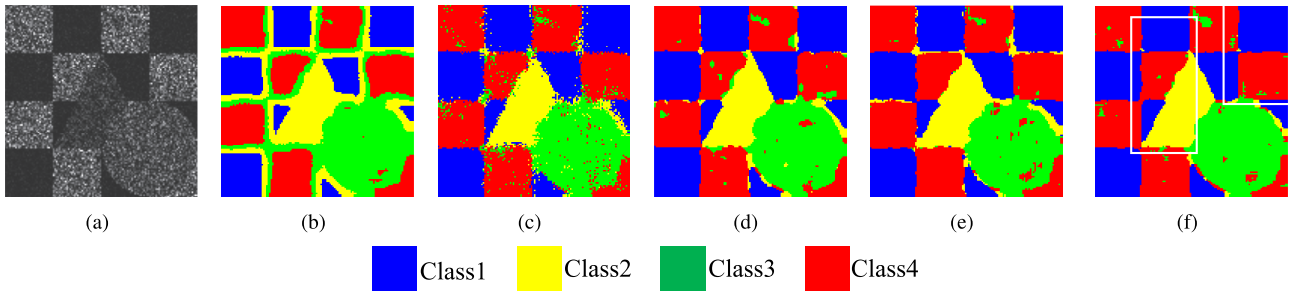

Fig. 12. Segmentation results of the simulated image. (a) 1-look simulated image. (b) Segmentation based on DBN model. (c) Segmentation based on SAE model. (d) Segmentation based on EPLS model. (e) Segmentation based on ACGAN. (f) Segmentation based on 3-DSM.

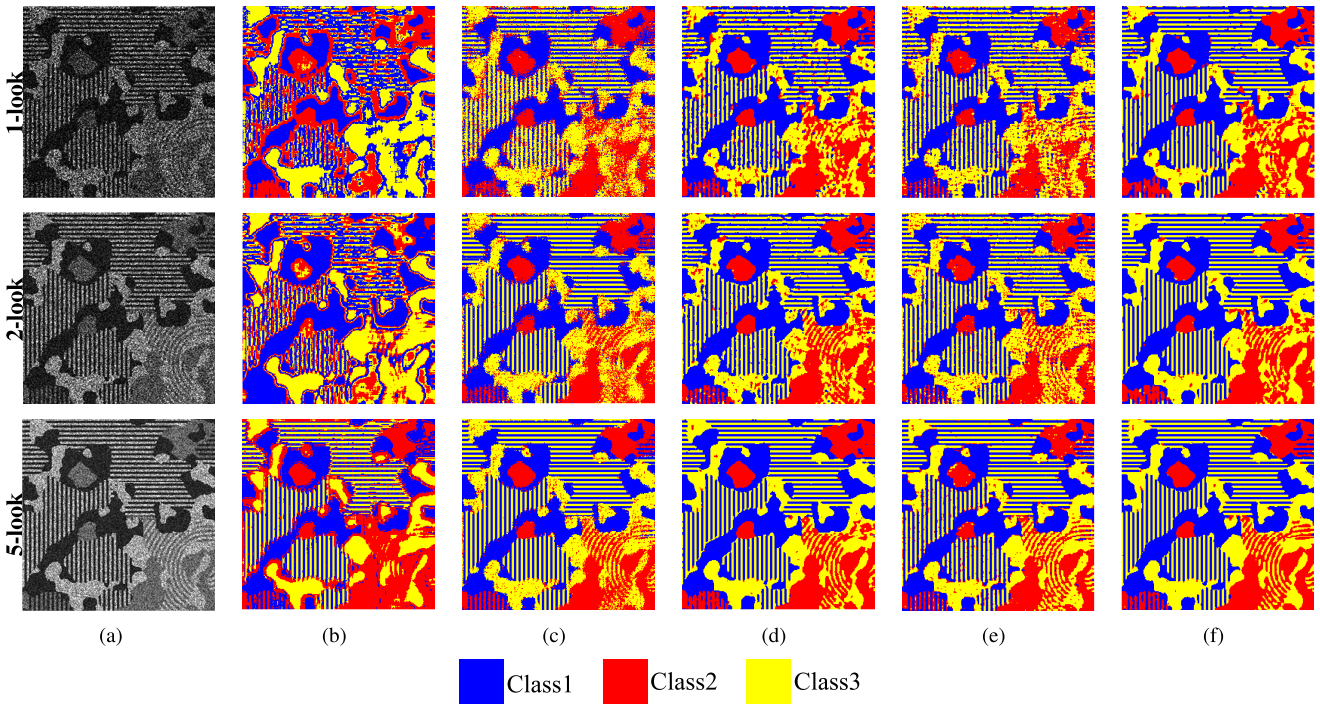

Fig. 13. Segmentation results of the simulated image set with different noise variances. (a) Simulated image set with ENL being 1-look, 2-look, and 5-look. (b) Segmentation based on DBN model. (c) Segmentation based on SAE model. (d) Segmentation based on EPLS model. (e) Segmentation based on ACGAN. (f) Segmentation based on 3-DSM.

[see Fig. 13(f)]. Concretely, as shown in Fig. 13(f), our model is more robust against speckle, resulting in smoother homogeneous regions. In addition, our model has a strong ability to capture details, and thus, the complex and fine stripe structures are segmented accurately. With the effective 3-D sparse maps for optimization, each convolution kernel is sparse and different. Thus, the extracted sparse features are more comprehensive, which is beneficial for classification. ACGAN is a supervised model requiring labels in training. The edges of Figs. 12(e) and 13(e) are not well preserved, and the results of ACGAN are greatly affected by the multiplicative noise. ACGAN is a relatively complex network and requires a large number of samples to train the parameters, while limited samples in our experiment restrict its performance.

Quantitatively, overall accuracy (OA) is used to evaluate the performance of each model, as listed in Table III. Overall, 3-DSM outperforms the other models. For the simulated images with different noise variances, the advantages of our 3-DSM are more obvious as the noise variance increases. Here, we focus on comparing EPLS with our model considering similar principles. In Fig. 13, the OA of 3-DSM is 3.23%, 2.19%, and 1.61% higher than that of EPLS model on the 1-look, 2-look, and 5-look simulated images. For the simulated image in Fig. 12, the OA of 3-DSM is improved by 1.31% over that of EPLS. Furthermore, the confusion matrices of EPLS and 3-DSM listed in Table IV are also used to validate our performance, where the row in the confusion matrix represents the actual label and the column denotes the

TABLE III
QUANTITATIVE EVALUATION OF THE SEGMENTATION FOR THE SIMULATED IMAGES

	DBN	SAE	EPLS	ACGAN	3-DSM
Fig.12	76.86%	91.19%	93.86%	93.35%	**95.17%**
Fig.13 1-look	56.67%	79.01%	86.94%	86.00%	**90.17%**
Fig.13 2-look	62.14%	86.91%	90.57%	89.09%	**92.76%**
Fig.13 5-look	75.01%	91.95%	93.73%	93.44%	**95.34%**

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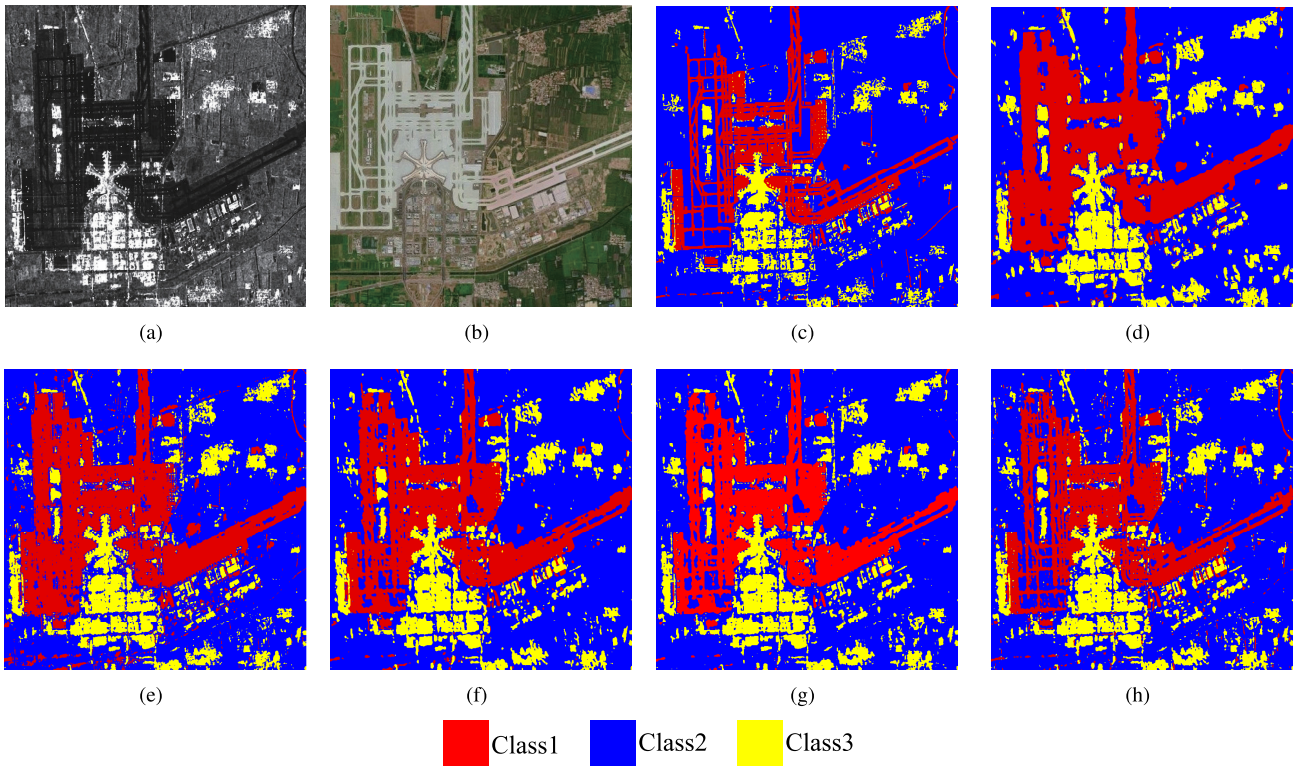

Fig. 14. Segmentation results of the real SAR image in the Daxing airport region. (a) Real SAR image of the Daxing airport region. (b) Google map. (c) Ground truth. (d) Segmentation based on DBN. (e) Segmentation based on SAE. (f) Segmentation based on EPLS. (g) Segmentation based on ACGAN. (h) Segmentation based on 3-DSM.

TABLE IV
CONFUSION MATRIXES OF EPLS AND 3-DSM MODEL

EPLS	Class1	Class2	Class3	Class4
Class1	**96.89%**	2.15%	0.18%	0.78%
Class2	0.59%	**95.80%**	2.10%	1.51%
Class3	0.12%	4.52%	**90.40%**	4.96%
Class4	0.67%	1.99%	4.89%	**92.45%**

3-DSM	Class1	Class2	Class3	Class4
Class1	**97.29%**	1.14%	0.22%	1.35%
Class2	0.39%	**97.24%**	1.25%	1.12%
Class3	0.72%	3.09%	**91.30%**	4.89%
Class4	0.72%	0.39%	4.12%	**94.77%**

predicted label. In particular, the segmentation accuracy of the 3-DSM enhances by 2.32% compared to EPLS for Class 4. On the other hand, compared to the EPLS model, our model has effectively reduced the misclassification of two visually similar categories. For example, there are 4.52% samples of Class 3 misclassified into Class 2 in the EPLS model, which is reduced to 3.09% in 3-DSM. The achievement can be attributed to the ability of 3-DSM that learns the independent sparse convolution kernels, which provides better perspectives for extracting more complete features.

F. Experimental Results on Real SAR Images

The real SAR images of Daxing airport in China, Oberpfaffenhofen and Noerdlinger regions in Germany, and Foulumn Test Site Area in Denmark are shown in Figs. 14(a)–17(a), respectively. The corresponding Google maps are shown in Figs. 14(b)–17(b). Ground truth maps are obtained based on the real SAR images and the Google maps. Daxing airport in Fig. 14(a) has complex structural information, Figs. 15(a) and 16(a) are urban areas with the complex scene, and Fig. 17(a) has many fine roads. Since these figures have complex scene and rich structural information, it is a challenging task to segment them. The segmentation results of DBN, SAE, EPLS, ACGAN, and our 3-DSM are shown in Figs. 14–17.

DBN and SAE are two common methods for unsupervised deep feature extraction, which attempt to reconstruct the input distribution to extract the features of the hidden layer. As shown in Figs. 14(d)–17(d) and 14(e)–17(e), we can find that DBN well maintains region homogeneity, and SAE preserves more structural information. However, the features learned by these methods are the side effects of reconstructing input rather than being explicitly explored. The sparsity and discrimination of features that are beneficial to segmentation are ignored. From the segmentation results in Figs. 14(d)–17(d) and 14(e)–17(e), we can find that DBN fails to preserve structural information, and SAE fails to maintain region homogeneity. The performance of the method modeling the input distribution is limited. Therefore, it is crucial to extract discriminative features for classification.

The EPLS model provides a new strategy for unsupervised parameter learning. Concretely, the model constructs the sparse matrixes with population sparsity and LS, which guides the learning of unsupervised representative features. EPLS moves the attention to the characteristic of features

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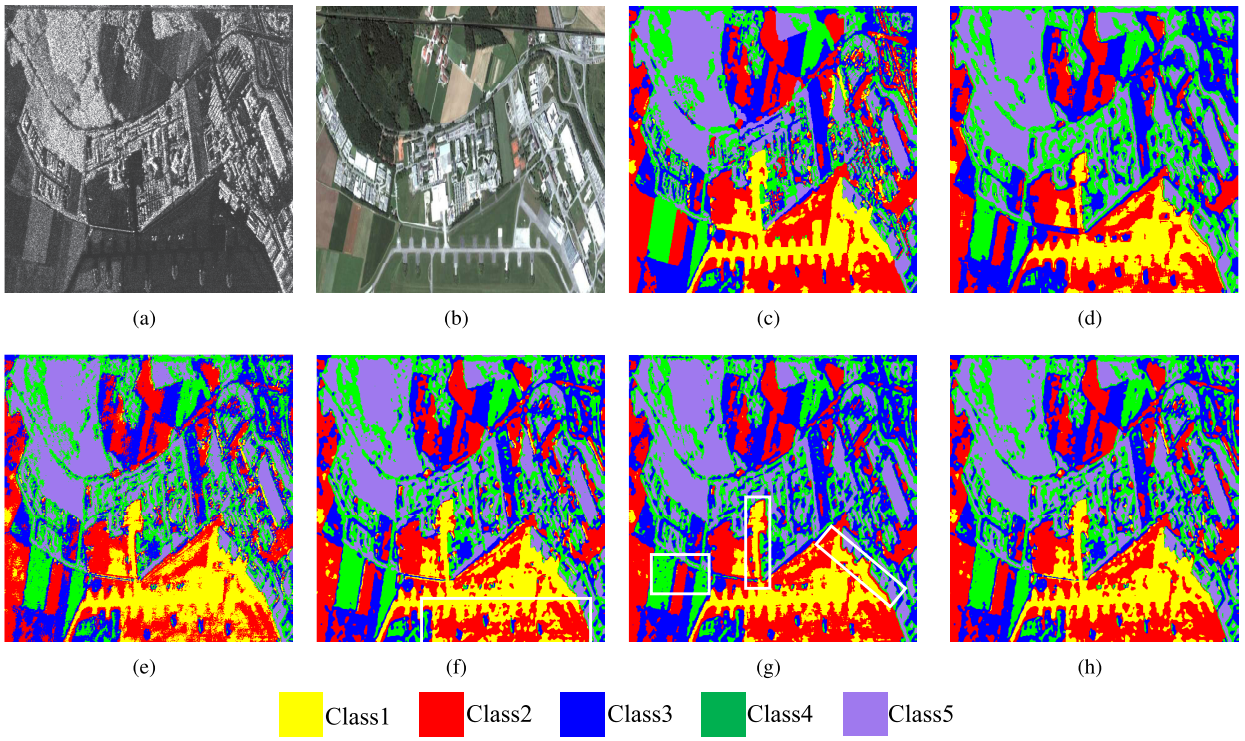

Fig. 15. Segmentation of the real SAR image in the Oberpfaffenhohen region. (a) Real SAR image of the Oberpfaffenhohen region. (b) Google map. (c) Ground truth. (d) Segmentation based on DBN. (e) Segmentation based on SAE. (f) Segmentation based on EPLS. (g) Segmentation based on ACGAN. (h) Segmentation based on 3-DSM.

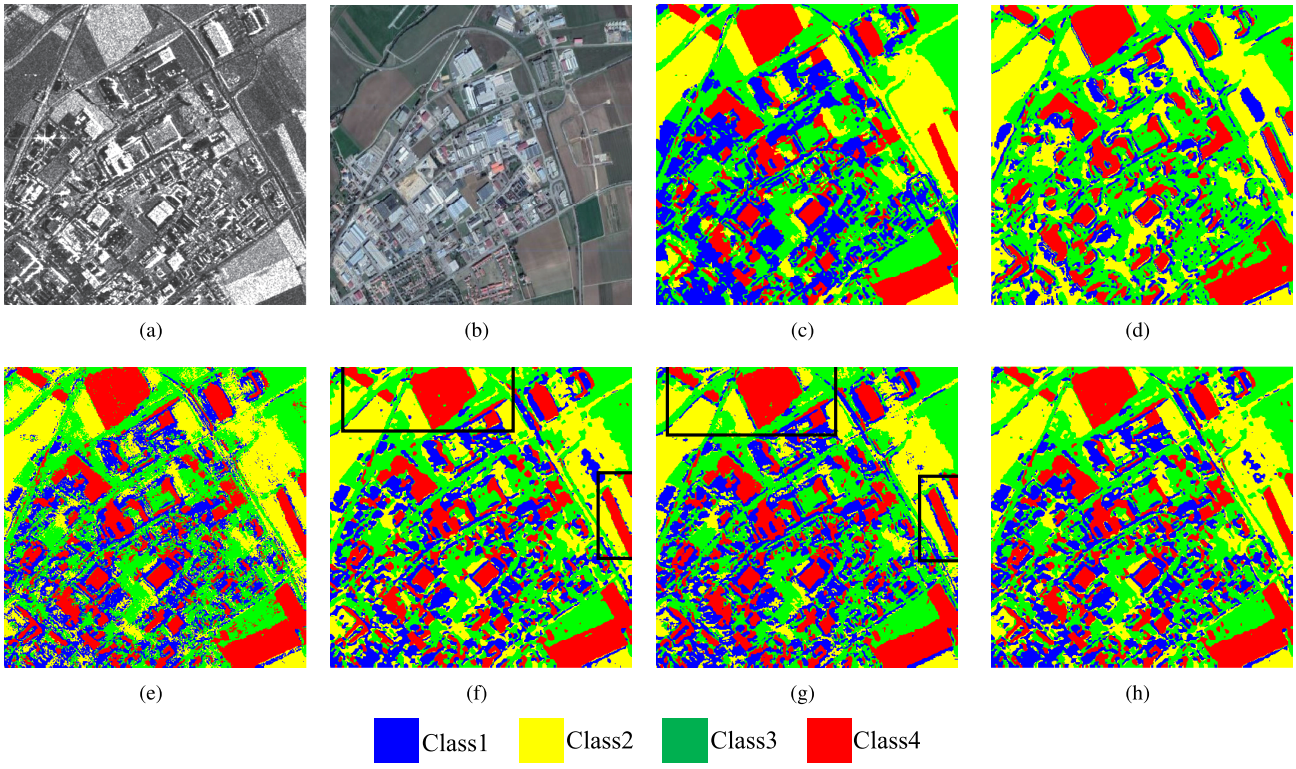

Fig. 16. Segmentation results of the real SAR image in the Noerdlinger region. (a) Real SAR image of the Noerdlinger region. (b) Google map. (c) Ground truth. (d) Segmentation based on DBN. (e) Segmentation based on SAE. (f) Segmentation based on EPLS. (g) Segmentation based on ACGAN (h) Segmentation based on 3-DSM.

and achieves significant improvement over DBN and SAE models. Visually, the segmentation results of EPLS model not only maintain a good homogeneous region but also preserve more structural information, as shown in Figs. 14(f)–17(f).

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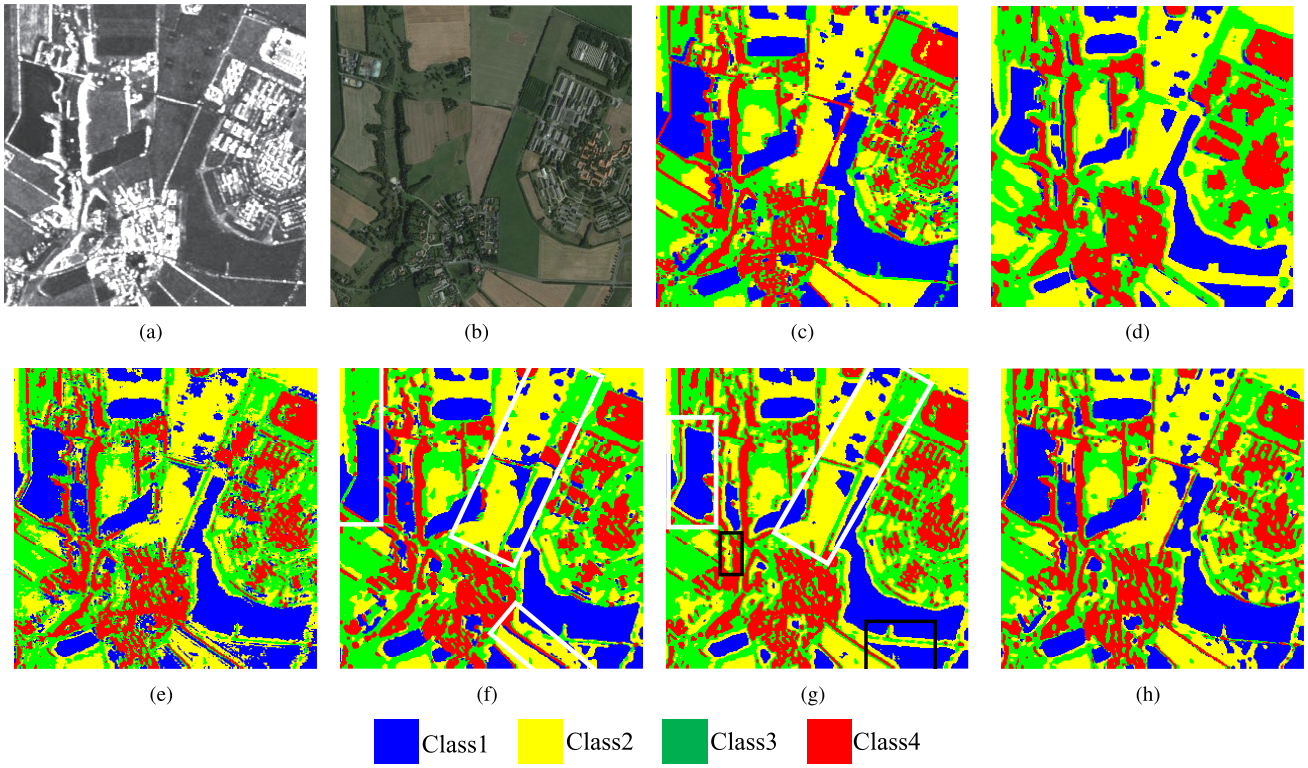

Fig. 17. Segmentation results of the real SAR image in the Foulumn region. (a) Real SAR image of the Foulumn region. (b) Google map. (c) Ground truth. (d) Segmentation based on DBN. (e) Segmentation based on SAE. (f) Segmentation based on EPLS. (g) Segmentation based on ACGAN. (h) Segmentation based on 3-DSM.

TABLE V

QUANTITATIVE EVALUATION OF THE SEGMENTATION FOR THE OBERPFAFFENHOFEN, NOERDLINGER, AND FOULUMN REGIONS

Region	DBN		SAE		EPLS		ACGAN		3-DSM	
	OA	Kappa	OA	Kappa	OA	Kappa	OA	Kappa	OA	Kappa
Oberpfaffenhofen	71.91%	0.6438	75.64%	0.6925	80.69%	0.7552	78.81%	0.7302	82.08%	0.7723
Noerdlinger	63.47%	0.5010	74.11%	0.6492	78.68%	0.7109	78.20%	0.7034	80.39%	0.7338
Foulumn	66.96%	0.5565	76.75%	0.6892	81.53%	0.7529	80.05%	0.7325	83.86%	0.7845

Quantitatively, as listed in Tables VI and V, the accuracies of EPLS model in the four regions are enhanced by 3.73%, 5.05%, 4.57%, and 4.78% compared to those by SAE. The task of EPLS model is to learn discriminative sparse features for classification. However, its sparse target is composed of a set of sparse vectors, which directly guides the learning of a set of feature vectors rather than the complete feature maps used for final classification. In this way, if the spatial information of the features is considered during training, the performance of the features could be further improved.

The proposed 3-DSM builds the 3-D sparse maps that have the same size as the feature maps as the optimization target. Concretely, the 3-D sparse maps are sparse in width, height, and depth, whose error with the corresponding feature maps constitutes the objective function. Thus, sparse convolution kernels can be learned in an explicit way. Since the principle of EPLS model and our 3-DSM is similar, we focus on comparing the performance of the EPLS model with that of our model. Besides, as the representation of supervised learning, the results of ACGAN are also compared together. First, our model has a strong discriminative ability to distinguish visually similar categories. Concretely, as shown in Fig. 14(a)–(c), Daxing airport is a large building with amounts of fine structures. Compared with DBN, SAE, EPLS, and ACGAN, our model precisely distinguishes the building (Class 1) from a similar background (Class 2) and maintains a lot of structural information. Similarly, EPLS performs poorly in visually similar categories, such as Classes 1 and 2 in the white pane regions of Figs. 15(f) and 17(f). Also, ACGAN could not distinguish visually similar Classes 1 and 2 in the black pane regions of Fig. 17(g) well due to the lack of training samples. Fortunately, the performance of our model outperforms the other models in these visually similar categories. In addition, our 3-DSM can locate the edge accurately. For example, the white pane regions in Fig. 15(g), the black pane regions in Fig. 16(f) and (g), and the white pane regions in Fig. 17(g) all have the issues where the edges of two different classes are classified to other classes. In comparison, 3-DSM can provide more accurate segmentation in these boundary regions with the complex scene. This is because each convolution kernel of 3-DSM is sparse and independent of each other, which provides a rich perspective for feature extraction. In this

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TABLE VI

QUANTITATIVE EVALUATION OF THE SEGMENTATION FOR THE DAXING AIRPORT REGION

Class	DBN	SAE	EPLS	ACGAN	3-DSM
Class1	87.45%	90.79%	92.63%	86.71%	**94.21%**
Class2	79.38%	78.93%	83.68%	85.50%	**85.90%**
Class3	89.31%	94.28%	94.16%	96.70%	**95.75%**
Overall accuracy	81.82%	82.61%	86.34%	86.90%	**88.38%**
Kappa	0.6407	0.6609	0.7248	0.7299	**0.7630**

explicit way, the extracted features not only achieve better sparsity and diversity but also have low redundancy. Therefore, 3-DSM has a strong ability to capture details, which can distinguish similar categories and accurately locate edges.

Quantitatively, Table V lists the evaluation results of the Oberpfaffenhofen, Noerdlinger, and Foulumn regions. For the three regions, the accuracies of 3-DSM are 1.39%, 1.71%, and 2.33% higher than those of EPLS model. Correspondingly, the kappa parameters of 3-DSM are 0.0171, 0.0229, and 0.0316 higher than those of EPLS. In particular, the accuracies of 3-DSM in the three regions are enhanced by 3.27%, 2.19%, and 3.81%, compared to those by ACGAN. ACGAN is the representative of supervised learning, which can realize data enhancement by the generator and feature extraction and classification by the discriminator. ACGAN requires sufficient training samples to learn the parameters of the generator and discriminator. However, in our experiment, training samples are limited. Furthermore, we list the detailed classification result of each model in the Daxing airport region, as listed in Table VI. Concretely, the accuracies of 3-DSM in the airport (Class1) and background (Class2) are 1.58% and 2.22% higher than those of EPLS model, respectively. ACGAN has poor performance in airport regions (Class 1) with the accuracy only being 86.71%, while our 3-DSM reaches 94.21% in these regions. This indicates that our model can well distinguish the background from the airport and has a strong ability to capture the structural information. This is consistent with the conclusion of the visual analysis. Finally, we can say that 3-DSM, as an unsupervised deep feature extracted method, is superior to DBN, SAE, EPLS, and ACGAN model for the segmentation of SAR images with the complex scene.

V. CONCLUSION

In this article, we propose an unsupervised 3-DSM to extract deep sparse features for the segmentation of SAR images with the complex scene. With the 3-D sparse maps being the optimization target, the sparse convolution kernels are learned in an unsupervised way. In particular, each convolution kernel is unique and has more perspective for feature extraction. The redundancy of the convolution kernels is reduced. Furthermore, each feature map possesses specific structural information. Thus, discriminative sparse features are extracted, which simplify and improve the representation of the input. Experimental results have proven that 3-DSM extracts sparse discriminative features with low redundancy and achieves better segmentation performance in maintaining structural information.

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